

1. Edge Impulse provides a good collection of learning blocks across different ML paradigms. In supervised learning, Classification is available for the prediction of categorical values (classification of motions into categories such as walking and idle), Regression can predict numerical data, such as flow rates from sensor data, and various Object Detection blocks are available, like FOMO for devices with limited processing capabilities and MobileNetV2 SSD FPN for devices with higher computational capacity. There are two types of Transfer Learning blocks for images and keyword spotting using a pretrained MobileNet model. In unsupervised learning, two anomaly detection blocks, K-means and GMM, are available, which are helpful in detecting abnormal values from sensor data. In addition, the Classical ML block supports logistic regression, random forests, XGBoost, and LightGBM.
2. The Neural Network Architecture panel in Classifier lets us use these layer types. The Input layer simply accepts input features without performing any calculations. Dense layers (fully connected) are fundamental units that connect each neuron to every neuron in the previous layer, enabling the learning of general mappings. Reshape modifies the shape of the tensor without changing its values, while Flatten flattens a multi-dimensional tensor into a 1D tensor for use by a Dense layer. Dropout randomly sets activation units to zero during training to help prevent overfitting. If the data is sequential, such as audio data, then 1D Convolution is used to learn local patterns by sliding a filter over the sequence, followed by 1D Pooling, which downsamples the output. Similarly, 2D Convolution and 2D Pooling are used for images and spectrogram data. Lastly, the Output layer is the final Dense layer, with an activation function of either softmax for classification or linear for regression.
3. The Deployment section gives us two important dimensions for compressing our model. The first dimension is the precision of the model: we can select either Unoptimized (float32) and retain all weights' precision or Quantized (int8) and compress the weights from 32-bit floating numbers to 8-bit integers via TensorFlow's Model Optimization Toolkit, which will reduce the size of the model by 4x while barely reducing its accuracy. The second dimension is the inference engine. For TFLite (LiteRT) on Microcontrollers, the standard interpreter loads the model into memory and executes it. On the other hand, the EON Compiler will compile the model into native C++ source code, thereby eliminating interpreter overhead and reducing memory usage by up to 55% in RAM and 35% in ROM without sacrificing accuracy. Finally, there is the option to use the EON Compiler (RAM Optimized), in which the model graph is sliced and computed as needed, resulting in an even greater reduction in RAM usage of up to 65% at the expense of increased latency.
4. The free plan provided by Edge Impulse includes the ability to generate synthetic data using three types of input: Images, Voice/Audio, and Time-Series data. To generate synthetic data from images, Edge Impulse can be connected to platforms like DALL-E to create labeled image samples generated from text prompts, which can be used to create image classification or object detection datasets without taking any pictures. To generate Voice and Audio samples, the Whisper model can be used to create speech samples

generated from text, while ElevenLabs can be used to generate audio samples like alarms or glass breaking. This way, one can easily create a dataset for applications like keyword spotting or sound classification.

Synthetic data can be used in machine learning since gathering real-world data is costly and time-consuming, while some edge cases may be hard or dangerous to record. Synthetic data can help address class imbalance and build a sample dataset before collecting real-world data. The downside of synthetic data is that it will likely not accurately represent the noise the sensor will experience on the target hardware; thus, it should preferably be used alongside real data.